

**Problem Set 6**  
Winter 2020

**Issued:** Monday, February 24

**Due:** Thursday, March 5

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Full credit: 140 (out of 180) points. Do problem 6.1 and a selection of your choice from the rest.

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**Problem 6.1** (30 points)

Let  $\Omega = \{1, \dots, m\}$  be a finite alphabet, and let  $\{\mathbb{P}_\theta, \theta \in \Omega\}$  be a probability model indexed by  $\Omega$ . Assume that we pick  $\Theta$  uniformly at random from  $\Omega$ . Given  $\Theta$ , an adversary picks  $k$  random letters from  $\Omega \setminus \{\Theta\}$  to form a set  $U = \{\Theta_1, \dots, \Theta_k\}$ . Then, the adversary draws  $X_1, X_2, \dots, X_n$  i.i.d. from the mixture distribution

$$\bar{P}_U := \frac{1}{k}(P_{\Theta_1} + \dots + P_{\Theta_k}).$$

The goal is to estimate  $\Theta$  based on the observation  $X^n := (X_1, \dots, X_n)$ . We would like to establish the following lower bound on the performance of any estimator  $\phi$ :

$$\mathbb{P}(\phi(X^n) \neq \Theta) \geq 1 - \frac{1}{m - k}. \quad (1)$$

- (a) First assume that we have access to  $U$ . Consider an estimator  $\psi$  that operates on  $U$  to produce an estimate of  $\Theta$ . Show that

$$\mathbb{P}(\psi(U) \neq \Theta) \geq 1 - \frac{1}{m - k}.$$

- (b) Assume that for three random variables  $\Theta, U$  and  $Z$ , we have the so-called Markov chain relation  $\Theta \rightarrow U \rightarrow Z$ , meaning that  $\Theta$  is independent of  $Z$ , given  $U$ . Show that

$$\min_{\phi} \mathbb{P}(\phi(Z) \neq \Theta) \geq \min_{\psi} \mathbb{P}(\psi(U) \neq \Theta).$$

This inequality can be thought of as the Bayesian decision-theoretic version of the *data-processing inequality* from the information theory.

- (c) Finish the proof of (1).

**Problem 6.2** (30 points)

Suppose that we know for  $X \sim \text{Poi}(\lambda)$  that

$$\mathbb{E}[X(X-1)] = \lambda^2, \quad \text{and} \quad \text{var}[X(X-1)] = 2\lambda^2 + 4\lambda^3.$$

Assume  $X_i \stackrel{\text{iid}}{\sim} \text{Poi}(\lambda)$ ,  $i = 1, \dots, n$ .

- (a) Let  $T_1 = \frac{1}{n} \sum_{i=1}^n X_i(X_i - 1)$ , argue that it is unbiased for  $\lambda^2$ , and compute its variance.

- (b) Let  $T_2 = \mathbb{E}[T_1 \mid S]$  where  $S = \sum_{i=1}^n X_i$ . Show that  $T_2 = S(S-1)/n^2$ .

*Hint:*  $(X_1, \dots, X_n)$  conditioned on  $S$  is distributed as a multinomial variable with parameters  $S$  and  $(1/n, \dots, 1/n)$ .

- (c) Show that

$$\text{var}(T_2) = \frac{4\lambda^3}{n} + \frac{2\lambda^2}{n^2}$$

*Hint:* The sum of independent Poisson variables is again Poisson distributed.

- (d) Conclude that Cramer–Rao bound for estimating  $\lambda^2$  can not be attained in this model.

**Problem 6.3** (15 points)

Keener 7.11: Consider a Bayesian model where  $\Theta \sim \text{Beta}(1, 2)$  and  $X_i | \Theta = \theta \stackrel{\text{iid}}{\sim} \text{Ber}(\theta)$  for  $i = 1, 2, \dots$ . Find  $\mathbb{P}(X_{n+1} = 1 | X_1, \dots, X_n)$ . This is often called Bayesian predictive distribution (or density).

*Hint:* Condition on  $\Theta$  and used your knowledge of the posterior  $\Theta | X_1, \dots, X_n$ .

**Problem 6.4** (20 points)

This problem considers minimax hypothesis testing. Let  $X \sim \text{Ber}(\theta)$  and the parameter space be  $\Omega = \{a, b\}$  for  $a = 1/3$  and  $b = 2/3$ . Take the loss to be  $L(\theta, \delta) = (\theta - \delta)^2$ . Any (nonrandomized) decision rule is specified by a pair of numbers  $(\delta_0, \delta_1)$  and any prior  $\pi$  by a single number  $\pi_a = \mathbb{P}(\Theta = a) \in [0, 1]$ .

- (a) Write down an expression for the frequentist  $R(\theta, \delta)$ .
- (b) Write down an expression for the Bayes risk  $r(\pi, \delta) = \mathbb{E}[R(\Theta, \delta)]$ .
- (c) Show that the Bayes rule is given by

$$\delta_0^* = \frac{2}{3(\pi_a + 1)} \quad \text{and} \quad \delta_1^* = \frac{4 - 3\pi_a}{6 - 3\pi_a}$$

- (d) Derive a necessary and sufficient condition for  $\delta$  to be an equalizer, i.e.,  $R(a, \delta) = R(b, \delta)$ .
- (e) Derive a minimax rule.

**Problem 6.5** (15 points)

This problem considers Bayesian multiple hypothesis testing. Let  $\Omega = \{0, 1, 2, \dots, m-1\}$ . Assume that  $X | \Theta = j \sim f_j(\cdot)$  for  $j \in \Omega$  where  $f_j(\cdot)$  is a density for each  $j$ . The prior on  $\Theta$  is specified as  $\pi_j := \mathbb{P}(\Theta = j)$ ,  $j \in \Omega$ . Consider the zero-one loss  $L(\theta, \delta) = 1\{\theta \neq \delta\}$ , that is,  $L(\theta, \delta) = 1$  if  $\theta \neq \delta$  and  $= 0$  otherwise.

- (a) Show that the Bayes rule is the Maximum a Posteriori (MAP) rule, i.e.

$$\delta_\Lambda(x) = \arg \max_{j \in \Omega} \mathbb{P}(\Theta = j | X = x) = \arg \max_{j \in \Omega} \pi_j f_j(x)$$

(b) For binary hypothesis testing where  $m = 2$ , show that the Bayes rule simplifies to

$$\delta_{\Lambda}(x) = 1 \left\{ \frac{f_1(x)}{f_0(x)} \geq \tau \right\}$$

for some suitable function  $\tau$  of  $\pi_0$  and  $\pi_1$ .

**Problem 6.6** (15 points)

Consider the Bayesian normal location family with normal prior on the location:

$$\begin{aligned} \Theta &\sim N(\mu, b^2) \\ X_1, \dots, X_n \mid \Theta = \theta &\stackrel{\text{iid}}{\sim} N(\theta, \sigma^2) \end{aligned} \tag{2}$$

We assume that  $\mu, b^2$  and  $\sigma^2$  are known.

(a) Argue that we can write  $X_i = \Theta + w_i$  where  $w_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$  independent of  $\Theta$ .

*Hint: Show that the conditional distribution of  $X_1, \dots, X_n$  given  $\Theta = \theta$  in this representation is the same as the one in (2)*

(b) Find the joint distribution of  $(\Theta, X_1, \dots, X_n)$ .

(c) Show that the conditional distribution of  $\Theta \mid X = x$  is

$$N((1 - \lambda_n)\bar{x} + \lambda_n\mu, 1/\tilde{\tau}_n^2)$$

where  $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$ , and  $\lambda_n$  and  $\tilde{\tau}_n^2$  that you specify.

**Problem 6.7** (15 points)

Let  $(X_1, \dots, X_n)$  be an i.i.d. sample from the mixture distribution with density

$$f_{\theta}(x) = \theta f_1(x) + (1 - \theta) f_2(x),$$

where  $f_i, i = 1, 2$  are two different known densities, and  $\theta \in (0, 1)$  is unknown.

(a) Show that the conditions

$$\frac{1}{n} \sum_{i=1}^n \frac{f_1(X_i)}{f_2(X_i)} > 1 \quad \text{and} \quad \frac{1}{n} \sum_{i=1}^n \frac{f_2(X_i)}{f_1(X_i)} > 1$$

are necessary and sufficient for the score equation (setting the derivative of the log likelihood to zero) to have a unique solution. Show that if there is a solution, then it is the MLE.

(b) Derive the MLE of  $\theta$  when the score equation has no solution.

**Problem 6.8** (20 points)

Given a fixed known integer  $r > 1$ , let  $X_{ij}, j = 1, \dots, r$  and  $i = 1, \dots, n$  be i.i.d. samples from  $N(\mu_i, \sigma^2)$ . Find the MLE of  $\theta = (\mu_1, \dots, \mu_r, \sigma^2)$ , and show that it is inconsistent for  $\sigma^2$  as  $n \rightarrow \infty$ .

**Problem 6.9** (20 points)

Suppose that  $X_1, \dots, X_n$  are i.i.d.  $N(0, \sigma^2)$ .

- (a) Show that  $\delta'(X) = \frac{C}{n} \sum_{i=1}^n |X_i|$  is a consistent estimator of  $\sigma$  if and only if  $C = \sqrt{\pi/2}$ .
- (b) Show that the MLE of  $\sigma$  is given by  $\delta_{MLE}(X) = \sqrt{\frac{1}{n} \sum_{i=1}^n X_i^2}$ . Determine the asymptotic relative efficiency of  $\delta'$  with  $C = \sqrt{\pi/2}$  compared to the MLE  $\delta_{MLE}$ .